

SUMMARY

Canadian Engineering graduate with hands-on experience in embedded systems, sustainable power generation, systems design. Driven to build intelligent, real world solutions where energy, infrastructure, and innovation meet.

EXPERIENCE

- University of Sharjah, Nuclear Research Assistant

Sept 2025 – Present

 - Developed python-based physics-informed AI agents & LLMs interfacing with Nuclear Power Plant (NPPs) simulations & physics engines
 - Participated in meetings with government agencies such as the Federal Authority for Nuclear Regulation (FANR) in the UAE to discuss technological innovations & the feasibility, safety, reliability, and methods of implementation for NPPs
 - Conducted stakeholder & public perception studies using qualitative analysis, structured elicitation, and decision-support tools (MCDM) following rigorous data-integrity requirements and adhering to established academic research standards.
- BricksSense, Embedded AI Circuits Design Engineer

Oct 2023 - Jan 2024

 - Designed analog cells and ran transistor- and chip-level simulations, solving design limitations creatively (Eagle), improving system efficiency.
 - Built Python-based AI automation systems for IoT home energy tracking, implementing PID control for energy usage monitoring and control, along with Selenium testing frameworks, reducing energy consumption through informing the end-user's decision making.
 - Maintained and validated embedded interfaces (STM32 firmware) in production systems, ensuring reliability and performance.
- CERC @ MARC, Undergraduate Research Assistant

May 2022 - Sept 2022

 - Collaborated in Dr. Ali Emadi's lab using Agile methods to develop V2X communication systems for intelligent traffic infrastructure interactions with self-driving vehicles, enhancing system usability.
 - Built full-stack Android apps (Kotlin/Java SDK) and backend APIs using MongoDB, Go, Python, and Firebase for journey reconstruction and optimization, improving journey efficiency.
 - Documented and deployed real-time data exchange pipelines across GPS, surveillance, and cellular systems (OpenCV), increasing data accuracy.
- Bank of Montreal, Process Specialist Co-op

May 2022 – Jan 2023

 - Managed internal software developmentz guidelines and tools including Jira within a Scrum framework, ensuring adherence to best practices and efficient project delivery in the bank's large, complex codespaces.
 - Documented and controlled software development processes using MS Office tools and VBA coding, enhancing process transparency and consistency.
 - Led the creation and rollout of a new internal training platform in Confluence, increasing employee engagement and knowledge retention.
- McMaster University Smart Freight Centre, Communications Assistant

Sept 2021 – Apr 2022

 - Published blog posts of interviews with people working on groundbreaking research in transportation and logistics using the latest technology.
 - Assisted in planning future in-person SFC events at McMaster.
- Royal Bank of Canada, Innovation Developer Co-op

Jun 2019 - Aug 2019

 - Developed ERP interfaces and backend logic using TypeScript, Angular, SQL, React Native, and NodeJS while utilizing unit and integration tests to enhance system performance and user satisfaction.
 - Utilized Git for version control, ensuring clean, reusable code standards and efficient collaboration with team members. Worked in Agile/Scrum workflows, contributing to timely and high-quality project deliveries.
- UpWork, Freelance (Technical Consultant, Full-Stack Developer)

2018 – Jan 2025

 - Developed and maintained full-stack web applications using Ruby on Rails, PostgreSQL, Docker, and React Native, enhancing system performance and user satisfaction
 - Led the upgrade and implementation of a redesign for Signals, a Vancouver media company, improving website responsiveness and cross-browser compatibility.
 - Utilized Git for version control, ensuring clean, reusable code standards and efficient collaboration with team members.
 - Created robust unit testing for Ruby on Rails applications using Rspec.

EDUCATION

- McMaster University B. Engineering Physics

2020 - 2025

Coursework included Nuclear Power Plant Systems and Operation, with training in reactor physics, fission chain reactions, CANDU reactor systems, reactor regulating and moderator systems, heat transport, turbine/steam cycles, and transient safety analysis. Designed and simulated nuclear plant control systems, including automation and PID logic for safety-critical infrastructure. Studied electrical power systems (SCADA, Smart Grids, PLC programming), and applied metrology methods for instrumentation, radiation measurement, and control system calibration. Additional background in solid-state materials and devices, quantum mechanics, nanomaterial characterization, and plasma physics for thermonuclear fusion. Strong foundation in machine learning.

Capstone Project: Developed a sustainable gravity-powered generator, awarded the \$1000 Joseph & Amy IP Award for innovation and feasibility.

LICENSES & CERTIFICATIONS

- IU Credential Ignition SCADA Software, Inductive Automation

Issued Aug 2025
- 1st in Innovative Design @ MEC 2023, Faculty of Engineering - McMaster University

Issued Nov 2023
- Applied Machine Learning With Python, University of Michigan

Issued Mar 2025

SKILLS

ITS & Control Systems: SCADA systems, PLC programming, OPC-UA, Modbus, Ethernet/IP, fiber optic networks, field devices (CCTV, VMS, sensors, controllers), alarm and historian configuration

Programming & Scripting: Python, Java, JavaScript, TypeScript, SQL, C/C++
Software & Tools: React, Angular, Node.js, Firebase, MongoDB, PostgreSQL, Docker, MATLAB/Simulink
Data & Visualization: SQL databases, reporting tools, data analysis (TensorFlow, scikit-learn), GIS basics
Collaboration: Git/GitHub, JIRA, Agile/SCRUM

HONORS & AWARDS

NSBE 50TH Annual Conference participant NSBE McMaster
Workshops facilitated by Apple, HP, US Department of Energy and more with engineers from all over North America
Graduate of Quantum Cryptography School for Young Students University of Waterloo
Participant in Middle East Energy 2025 Conference
McMaster Engineering Competition 1st Place winner Joseph and Amy IP student award awarded to students who represent McMaster University at the OEC through exceptional Engineering Physics Capstone projects

LEADERSHIP & EXTRACURRICULAR

Hack Club, Community Leader Jan 2018 - Aug 2020
• Developed tools using JavaScript, React, and Python to support global student dev communities, ensuring cross-browser compatibility and accessibility.
• Organized, hosted, and participated in events with tech leaders (e.g., Elon Musk, Alex Stamos, Limor Fried) to promote inclusive STEM engagement.
• Worked closely with COO, Co-founder, and Board Member Christina Asquith to increase community engagement and equitable opportunities for high school girls interested in technology. Achieved 100% increase (double) in female participation at Hack Club's workshops
• Pitched for funding to Craig Newmark, Craigslist founder, for an initiative to distribute Macbooks to young girls in need of resources worldwide. Upon approval of the project, I led and organized the distribution. Achieved 40% of 1st round grants distribution to young women in need
• Gathered and presented diversity statistics on community and their participation, coordinated collaboration with Girl Genius for 6 events/workshops.

Public Lab Plots 2, Contributor Oct. 2018
• Used Ruby on Rails along with JS to add features and thoroughly test a website's front and back end for Public Lab, a non-profit where data collected by hobbyists and scientists is shared so communities can easily track local pollution levels and set goals to reduce them.

Rebelle Zine, Web Design Volunteer 2022 - Present
• Implementing and consulting on redesign of the Rebelle Zine (online magazine for women-led article writing, businesses, creative pursuits, and the arts) using SquareSpace, Webflow, CSS/SASS, Figma UI/UX design

TechNova, Workshop Mentor and Panelist Speaker Aug 27 – 29 2021
• Helped students work through Figma UI/UX and a React JS workshop at the University of Waterloo’s TechNova hackathon. Spoke as a panelist at the University Living and Everything Internships Panels during the event.